

## **Exp No: 25**

### **PRODUCER–CONSUMER PROBLEM USING SYSTEMC**

#### **AIM:**

To implement and simulate the Producer–Consumer problem using SystemC.

#### **PROGRAM:**

```
#include <systemc.h>

// Producer Module
SC_MODULE(Producer) {
    sc_fifo_out<int> out;

    void produce() {
        for (int i = 1; i <= 5; i++) {
            cout << "Produced: " << i << endl;
            out.write(i);
            wait(1, SC_SEC);
        }
    }

    SC_CTOR(Producer) {
        SC_THREAD(produce);
    }
};

// Consumer Module
SC_MODULE(Consumer) {
    sc_fifo_in<int> in;

    void consume() {
        int value;
        while (true) {
            in.read(value);
            cout << "Consumed: " << value << endl;
```

```
wait(1, SC_SEC);  
}  
}
```

```
SC_CTOR(Consumer) {  
    SC_THREAD(consume);  
}  
};
```

```
int sc_main(int argc, char* argv[]) {  
    sc_fifo<int> fifo(5);
```

```
    Producer producer("Producer");  
    Consumer consumer("Consumer");
```

```
    producer.out(fifo);  
    consumer.in(fifo);  
    sc_start(10, SC_SEC);
```

```
return 0;

}
```

